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To cite this article: Ruwini Edirisinghe & Lahiru Gunathilake (2025) IoT-based smart vest for heat stress management in construction, International Journal of Construction Management, 25:16, 2137-2149, DOI: [10.1080/15623599.2025.2504550](https://doi.org/10.1080/15623599.2025.2504550)

To link to this article: <https://doi.org/10.1080/15623599.2025.2504550>



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Published online: 17 May 2025.



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IoT-based smart vest for heat stress management in construction

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ABSTRACT

Heat stress is a major challenge in construction, affecting workers' safety and productivity. However, existing systems often lack real-time monitoring capabilities, personalization features, and power efficiency, limiting their effectiveness. To overcome these limitations, this work presents an Internet of Things (IoT)-based smart vest, specifically designed for the construction sector. The experimental development of an IoT-based circuit followed rigorous circuit design guidelines and testing protocols. The components and sensors in the circuit were carefully designed, including IoT capabilities to cater for the construction sector's hostile work and environmental conditions. Workers' health status will be continuously monitored without disrupting regular work. The proposed system demonstrated strong performance, with an error range of -0.4 to 0.06°C and continuous operation of data transmission. The developed IoT-based smart vest allows for seamless communication with cloud and local smart devices, enabling global access and effective real-time data visualization and warnings for workers and management. Temperature is used as the primary metric. Future research by the authors will incorporate heart rate variability, galvanic skin response, and humidity to enhance heat stress assessment. Sensor accuracy and reliability were validated with the potential for integration into construction safety protocols to enable a proactive response to heat stress management.

ARTICLE HISTORY

Received 8 December 2024
Accepted 3 May 2025

KEYWORDS

Wearables; IoT; smart PPE;
heat stress; construction
management; smart vest

Introduction

The construction industry has been found to be more susceptible to heat stress compared to other industries (Acharya et al. 2018; Edirisinghe and Andamon 2019; Borg et al. 2021). Construction workers are subjected to heat stress during outdoor physical work on extremely warm days, and working in confined spaces under these hot conditions could result in even more severe health problems (Umar and Egbu 2020; Karthick et al. 2023). Further, global costs from lost worktime were US\$280 billion in 1995, 311 billion in 2010, and are projected to reach 2.4–2.5 trillion in 2030, and up to 4 trillion by 2100 (Borg et al. 2021). Therefore, prevention of heat stress among construction workers is critically important. Despite the existence of traditional methodologies such as physically monitoring workers for any signs or symptoms of heat illness, wearing light-coloured, breathable clothing, adjusting work schedules based on heat conditions, and promoting hydration, there remains a gap in technology-based solutions for real-time monitoring and validity of workers' health status (Karthick

et al. 2023), while (Cannady et al. 2024) highlight limitations including the exclusion of relevant studies due to database selection, the lack of quality assessment, and the absence of direct validation research in real-world occupational settings.

The digitalisation of construction sites offers opportunities for pervasive, context-aware environments equipped with intelligent systems where future construction workers will wear IoT-based smart personal protective equipment (PPE) on site. For example, construction workers who are part of the “digital skin” (Edirisinghe 2019) of future construction site will wear smart wearables such as e-textiles, smart safety vests, smart hard hats, smart glasses, and even an upper body exoskeleton. Similarly, Forcael et al. (2020) highlight the pivotal role of key technologies such as IoT, big data, and virtual reality in driving the paradigm shift towards industrial transformation.

Recognising the need to address challenges in heat stress management, this paper introduces an experimental setup of an IoT-based smart vest tailored to measuring the micro-climate and skin temperature of construction workers. By harnessing IoT technology,

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this novel solution enables continuous monitoring, facilitating the timely detection of heat stress indicators and proactive intervention to mitigate risks. This study aims to validate the accuracy, reliability, and feasibility of a smart vest-based heat stress management system, which utilizes a temperature sensor for real-time monitoring. Specifically, the sensor's accuracy is evaluated in laboratory setup to detect heat stress conditions, its practical usability in high-temperature workplaces, and its potential advantages over traditional methods such as manual observation and work-rest schedules. It is hypothesized that the smart vest's sensor, which has shown high accuracy in laboratory settings, will provide a reliable, scalable, and effective solution for heat stress monitoring in occupational environments.

The rest of the paper is organised as follows: the literature review offers an overview of research on innovative technologies for heat stress management in the construction industry. The methodology section details the theoretical underpinning, design and development steps. The results and discussion provide a comprehensive exploration of technical validation and practical implementation strategies for managing heat stress in construction prior to the paper's conclusion.

Literature review

Plethora of studies argue for the limited effectiveness of traditional heat stress management strategies in construction (Edirisinghe and Andamon 2019). This section provides a comprehensive exploration of innovative technologies.

Applications of wearables

Wearable devices have evolved to offer multifaceted capabilities across various fields. For example, recent advancement of wearables in healthcare are flexible miniaturised sensors (FMS) in gathering health-related data for personalised healthcare (He et al. (2022) and breathable electronic skin (e-skin) tailored for daily physiological monitoring (Yang et al. (2022). Further advancements in wearable technologies are evident in healthcare applications, where Mukhopadhyay et al. (2022) provide insights into the deployment of wearable sensors for monitoring physiological movements and signals. Beyond healthcare, Jin et al. (2022) propose leveraging edge computing to enhance the capabilities and convenience of wearable devices, addressing limitations stemming from their lightweight design. Moreover, Trovato et al. (2022) highlight the demand for miniaturised sensors, particularly in medical devices embedded into textiles, showcasing their

potential for smart health monitoring. This comprehensive overview underscores the diverse applications and promising future of wearable technologies across various domains, including construction, sports (Chidambaram et al. 2022), machine learning integration (Dai et al. 2023), affordable health research (Huhn et al. 2022), and early detection of COVID-19 (Hui et al. 2022).

In mining, Dambe et al. (2023) developed systems to monitor environmental conditions and vital signs, facilitating swift responses to accidents. Xiao and He (2024), developing smart vests equipped with flexible pressure sensors for real-time respiratory rate monitoring in sports and medical care settings. In construction safety, studies explore factors influencing the adoption of wearable technologies (Chong et al. 2023) and integrated metal detectors into safety vests to enhance workplace safety ((Rajendran et al. 2020).

Moreover, Sharma et al. (2022) introduced a sensor-based safety helmet capable of monitoring environmental variables, heat stress indices, and physiological parameters, demonstrating its efficacy in different work environments. Pancardo et al. (2015) proposed real-time personalised monitoring for occupational heat stress estimation, leveraging data from environmental and body sensors to enable timely notice for workers to control heat stress. The first e-textile-based application, a prototype piece of smart clothing, was presented by Edirisinghe and Blismas (2015). That prototype sensed temperature abnormalities and alerted wearers and surrounding workers or management. These innovative solutions not only enhance workplace safety but also empower workers and employers to effectively manage heat stress exposures in high-risk environments.

The integration of wearable technologies with IoT capabilities has the potential to overcome these limitations by enabling continuous, real-time data collection and remote monitoring. As a result, researchers and industry professionals are increasingly focusing on IoT-based solutions for more effective heat stress management in construction.

Harnessing IoT for heat stress management in construction

The literature includes some studies or prototypes of IoT-based wearables despite the limited focus on applications tailor-made for construction sites. An alerting system, indicating significant increases in body temperature in real-time, aimed at preventing heatstroke in children was introduced by Zahari et al. (2022). A wearable IoT-based heatstroke detection device introduced by Son et al. (2021) demonstrated

Table 1. Literature summary.

Research/Product	Significance	Research gaps
(Kim et al. 2020)	IoT-based construction worker physiological data monitoring platform using wearable smart bands, which integrated an overall heat assessment (OHS) and a personal management system (PMS)	Only approved personnel can monitor the status and make necessary decisions remotely
(Costin et al. 2019)	IoT-based active leading indicator (ALI) system that automates the collection of real-time data to identify and respond to safety hazards	Future work should focus on addressing privacy concerns, securing proper authorizations for collecting personal data to enhance scientific analysis, and expanding the system to cover additional safety incidents. Additionally, future research should evaluate hardware limitations, develop hybrid connectivity solutions for remote sites, and consider the unintended consequences of IoT-enabled wearable sensors, such as behavioural changes, data privacy, and regulatory challenges.
(Ojha et al. 2024)	A worker-centred heat strain monitoring framework that integrates wearable biosensors, ensemble learning, and domain adaptation, achieving over 93% prediction accuracy, and advancing real-time heat risk assessments and early warning systems for heat-related fatalities in diverse work environments	The study's limitations include the use of commercially available biosensors with restricted physiological measurements, the exclusion of factors like hydration and acclimation, and the need for further validation of the model in real-world environments with more comprehensive bio-signal monitoring and proactive safety interventions.
(Sergi et al. 2021)	Smart system leveraging wearable devices and IoT technologies to monitor physiological strain responses and to detect heat-stress conditions in industrial indoor environments in real time with low-energy Bluetooth	Only focused on the thermal comfort of indoor environments

effective detection of potential heatstroke by analysing the thermal heatstroke risk coefficient (THSRC), core body temperature, and heart rate, providing early alerts to users in hot environments. An IoT heatstroke shield developed by Javed et al. (2020) addresses the significant problem of heatstroke using a wristband and alert system to monitor and notify wearers of dangerous body parameter readings. Furthermore, Tamura et al. (2022) introduced an IoT system for heatstroke prevention, incorporating real-time monitoring of deep body temperature using a dual-heat-flux thermometer which can alert users to potential heatstroke during sudden heat waves. The development of a smart real-time monitoring system using IoT technology, integrating sensors for heat stress and toxic gas detection was presented by Jamal et al. (2019). The system demonstrates high accuracy in measuring toxic gases, temperature humidity index (THI), and wind chill index (WCI), providing valuable insights for mitigating health risks associated with temperature extremes. An IoT-based construction worker physiological data monitoring platform using wearable smart bands was introduced by Kim et al. (2020), integrating an overall heat assessment (OHS) and a personal management system (PMS). Similarly, Sergi et al. (2021) proposed a smart system leveraging wearable devices and IoT technologies to monitor physiological strain responses and detect real-time heat-stress conditions in industrial environments with low-energy Bluetooth.

So-far limited research focus on IoT-based solutions for heat stress management offers promising

avenues for enhancing workplace safety and protecting worker health on construction sites. The laboratory evaluation of thermal comfort of a smart safety vest presented by Edirisinghe and Jadhav (2017), simulated working conditions and metabolic rates of construction workers using thermal mannequins. Later, Edirisinghe et al. (2022) presented a holistic heat stress management system trialled on construction sites in real time.

Table 1 provides a summary of recent implementations, highlighting the significance and the research gaps.

The literature underscores the diverse applications of wearable technologies across healthcare, sports, and industrial sectors, with significant potential for construction safety. While early research has explored wearable heat stress management solutions, the transition to IoT-based approaches remains an evolving field. Current studies demonstrate the feasibility of real-time physiological monitoring and environmental hazard detection, yet construction-specific IoT applications require further development. This research aims to bridge that gap by developing a smart vest tailored for temperature monitoring in construction settings, leveraging IoT capabilities to enhance worker safety and mitigate heat-related risks.

Methodology

In this experimental study, the primary objective was to develop tailor-made personal protective equipment (PPE) that is water-resistant, lightweight, and IoT-enabled. By following this structured approach, the

proposed system progressed through various stages of development, from initial concept and feasibility studies to prototype testing and final validation in real-world conditions. This process guarantees that the system is not only innovative but also reliable and ready for deployment in practical applications.

The foundational principles of circuit design were derived from established theoretical frameworks and empirical studies in the field of electrical engineering (Warren 1989; Pecht 1994; Gallagher and Lawlor-Wright 1995). The flow diagram in Figure 1 illustrates the overall methodology.

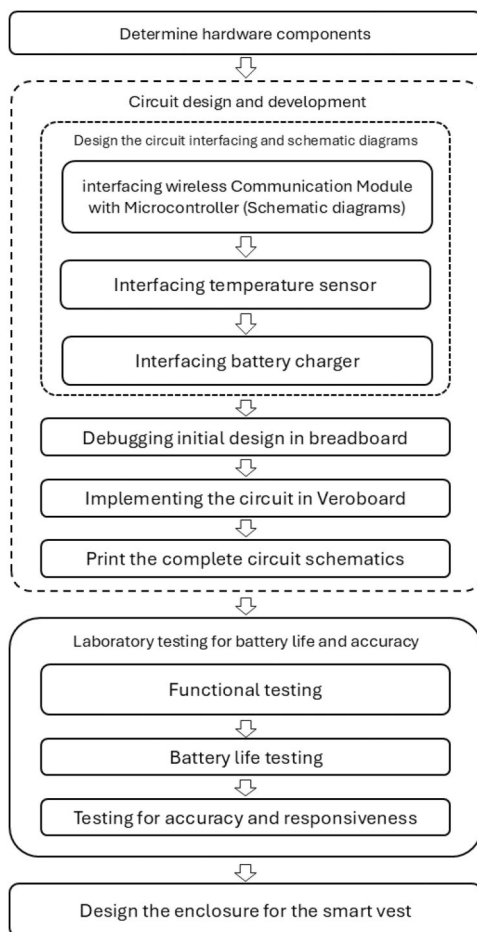


Figure 1. Flow diagram of methodology.

Smart vest circuit design and development

The proposed IoT-based smart vest consists of a circuit with several modules such as temperature sensing and reading, battery charging, and a wireless transmission system to directly communicate the data to a webserver. Figure 2, below, shows the connections of the system modules and the external cloud. System design and modules are discussed in detail below.

Main control unit

The main control unit, as the name implies, is the heart of the system that controls all the other sub-units. This consists of a Microchip PIC24FV family microcontroller (PIC24F16KA301.301 2024) to provide all the system functionality including low power consumption, real-time processing capabilities, and reliability in embedded applications. The PIC24FV16KA301 operates with an active current of 2.9 mA/MHz and can reduce power consumption to as low as 100 nA in deep sleep mode, making it highly suitable for energy-efficient applications. While the ESP32 offers integrated Wi-Fi, its higher power consumption (160–260 mA active, $\sim 10 \mu\text{A}$ deep sleep) and increased complexity made it less suitable for this application (ESP32 2024). PIC24FV16KA301 devices provide CPU performance up to 16 MIPS at 32 MHz and consist of an 8 MHz internal oscillator with 4 x PLL option and multiple clock frequency divide options. The device peripherals consist of multiple serial communication modules, including two SPI modules, two I2C modules and two UART modules supporting RS-485 and RS-232. In this design, the latter is used for serial communication and debugging. This family of PIC microcontrollers provides the ability to manage power consumption by selectively managing clocking to the CPU and peripheral devices. Also, the PIC24F devices support a wide range of clock frequencies allowing switching between low-power or high-precision oscillators to manage power consumption. Moreover, PIC24F devices give additional functionality by providing instruction-based power-saving modes, which optimises the power usage of the connected system.

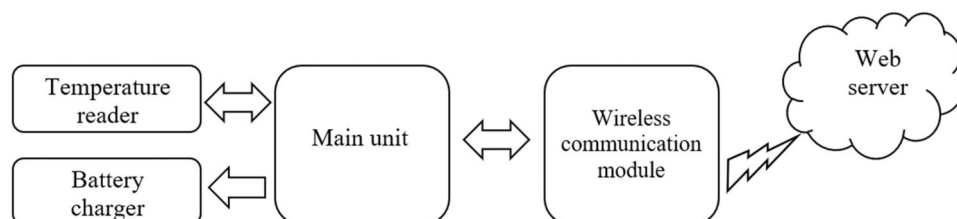


Figure 2. System design block diagram of the smart vest and its external connectivity.

A PIC24FV16KA301 device with 20 pins, 16Kb of programmable memory and other available peripherals accomplish the minimum requirements for the design, enabling its functionality with nominal performance. The following circuit diagram shows the basic connections to incorporate the PIC24FV16KA301 into the design.

The system firmware is developed in MPLAB X IDE v3.20 on Windows 7 and the PIC24FV16KA301 device is programmed with a simple, low-cost in-circuit PICKit3 programmer/debugger (PICKIT3.3 2024). The PICKit 3 programmer/debugger is a debugger system used for hardware and software development of Microchip PIC microcontroller units (MCUs).

Firmware configuration bits are set to enable clock switching and fail-safe clock monitor and 8 MHz Fast RC Oscillator with post-scalar. The peripheral clock frequency of 32 MHz is generated by using the device's 8 MHz internal oscillator with post-scalar 1 and PLL module selected.

Wireless communication module

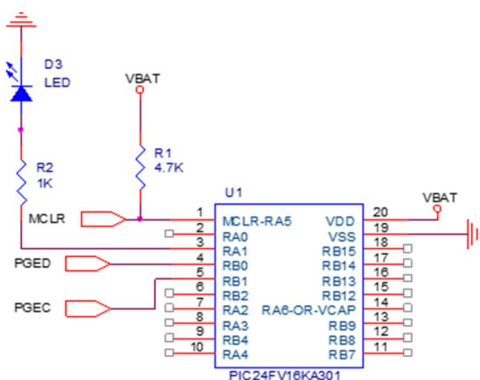
A low-cost ESP8266 wireless module (ESP82668266 2024) is interfaced with the PIC24FV16KA301, as shown in Figure 3(a) to enable wireless communication of temperature data with a web server. The ESP8266 was selected for its cost-effective and efficient connectivity, meeting the system's communication requirements while consuming less power than the ESP32. It operates at $\sim 70\text{--}200\text{ mA}$ during transmission and $\sim 20\text{ }\mu\text{A}$ in deep sleep mode, whereas the ESP32 consumes $\sim 160\text{--}260\text{ mA}$ when active and $\sim 10\text{ }\mu\text{A}$ in deep sleep, making the ESP8266 a more energy-efficient choice for this application (ESP32 2024; ESP82668266 2024). The ESP8266 module supports the IEEE 802.11 b/g/n protocol and has an

integrated TCP/IP protocol stack. The ESP8266 provides excellent functionality in mobile, wearable electronics and IoT applications with low power consumption. The module can be operated in three modes: Standalone (STA), Access Point (AP), and STA + AP, enabling self-contained Wi-Fi networking by hosting the application or offloading all Wi-Fi networking functions from another application processor. The ESP8266 provides wireless internet access to PIC24FV16KA301-based designs *via* serial communication through the microcontroller's UART interface at a Baud Rate of 115,200bps. The circuit diagram in Figure 3(b) shows the interconnection between the ESP8266 module and the PIC24FV16KA301.

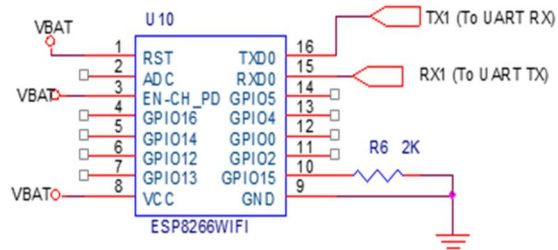
An ESP8266 Wi-Fi module is configured *via* AT commands to operate in Standalone mode, allowing multiple TCP connections. The system firmware is developed to initiate the connectivity with a wireless network by communicating AT commands *via* the ESP8266 module. A TCP connection is initiated with the webserver and the measured temperature reading is transmitted to the server using the channel API key.

Interfacing the wi-fi module with the micro controller

The device's full-duplex UART1 serial I/O module is enabled and configured to communicate with the ESP8266 Wi-Fi Module with a Baud Rate of 115,200 bps. The UART1 MODE register is configured to activate UART1 with High Baud Rate enabled. The UART module includes a dedicated 16-bit Baud Rate Generator (BRG). The U1BRG value of 34 is calculated by applying Instruction Clock Frequency $FCY = (FOSC/2) = 16\text{ MHz}$ and the required Baud Rate of 115,200 bps in Equation 1.



(a) Schematic – PIC24FV16KA301 Basic Connections



(b) ESP8266 connections with PIC24FV16KA301 UART TX and RX pins

Figure 3. Schematic diagrams PIC24FV16KA301 connections.

$$U1BRG = \frac{FCY}{4 \times \text{Baud Rate}} - 1 \quad (1)$$

The calculated Baud Rate can be obtained by back-substituting U1BRG value 34 in the above equation and the percentage error can be obtained as in Equation 2.

$$\text{Error} = \frac{(\text{Calculated Baud Rate} - \text{Desired Baud Rate})}{\text{Desired Baud Rate}} \times 100\% \quad (2)$$

Therefore, in the system implemented, an 8 MHz internal oscillator with post-scalar 1 and PLL module is selected with U1BRG value 34 to achieve 115,200 bps High Baud Rate with a maximum 0.8% Baud Rate error.

The ESP8266 Wi-Fi module is configured *via* AT commands to operate in Standalone mode, allowing multiple TCP connections. The system firmware is developed to initiate the connectivity with a Wireless network by communicating AT commands with the ESP8266 module.

Interfacing the temperature sensor with the microcontroller

The temperature sensor used in the system is a DS18B20 digital thermometer (DS18B20.20 2024), which provides 9-bit to 12-bit Celsius temperature measurements. The DS18B20 temperature sensor was selected due to its accuracy, simplicity, and ability to operate in a low-power mode around 1.5 mA current with minimal wiring. The DS18B20 communicates over a 1-wire bus that requires only one data line for communication with the central microcontroller. The sensor operates from -55 to $+125^{\circ}\text{C}$ and is accurate to $\pm 0.5^{\circ}\text{C}$ across the range from -10° to $+85^{\circ}\text{C}$. Moreover, the DS18B20 has the ability to derive power directly from the data line, eliminating the need for an external power supply. The system uses a waterproofed version of the DS18B20 sensor, allowing accurate temperature measurements in wet conditions. Using a digital sensor avoids signal degradation over long distances, which is useful in measuring the temperature in a specific area in the body away from the main system. Each sensor has a unique 64-bit serial number, allowing for multiple sensors to be used on one data bus. The circuit diagram in Figure 5 (a) shows the connection of the DS18B20 with the PIC24FV16KA301. The firmware is developed to read the digital temperature value given by the sensor *via* a one-wire protocol implementing read-write timing as specified in the DS18B20 datasheet. The obtained 16-bit temperature data are manipulated bitwise in order to extract the corresponding temperature at the correct resolution with the sign.

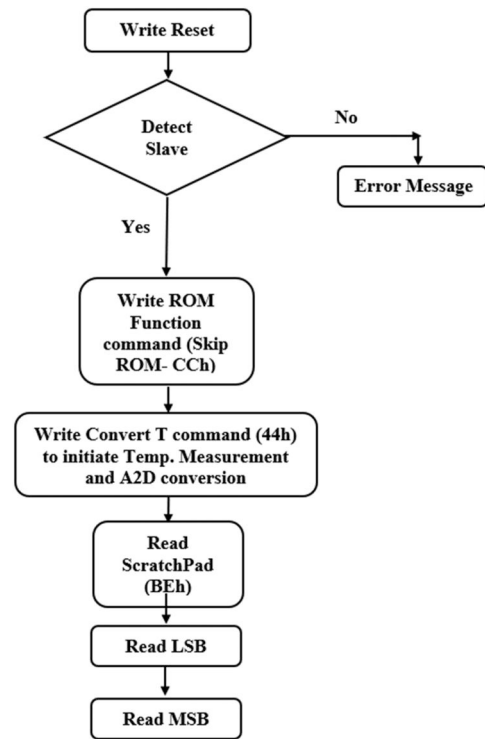


Figure 4. Instruction flow diagram for reading temperature value *via* one-wire protocol.

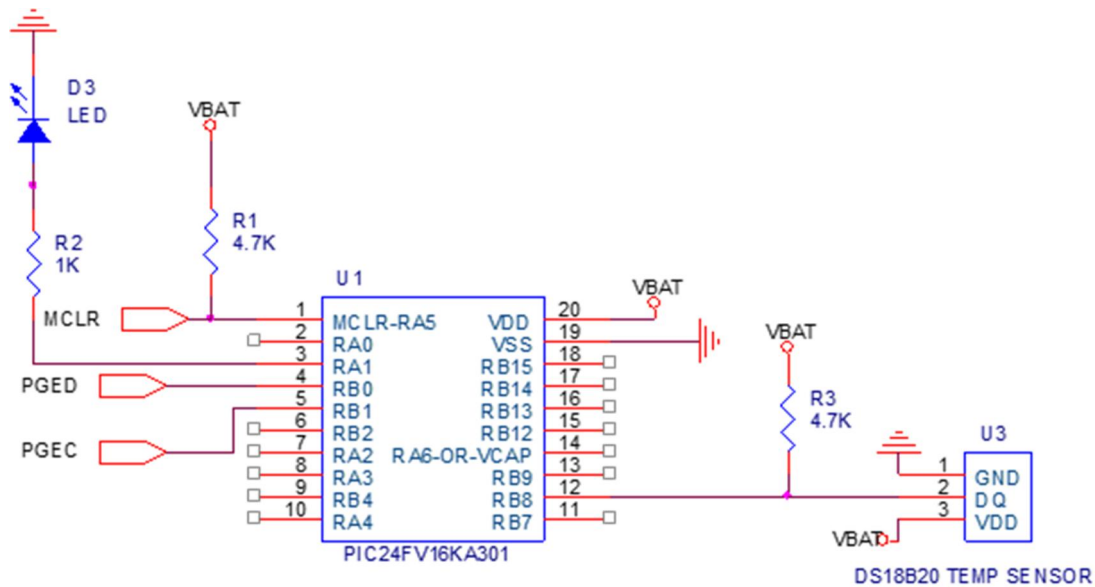
The block diagram in Figure 4 provides the algorithm for reading a temperature value from the sensor.

Interfacing with the battery charger

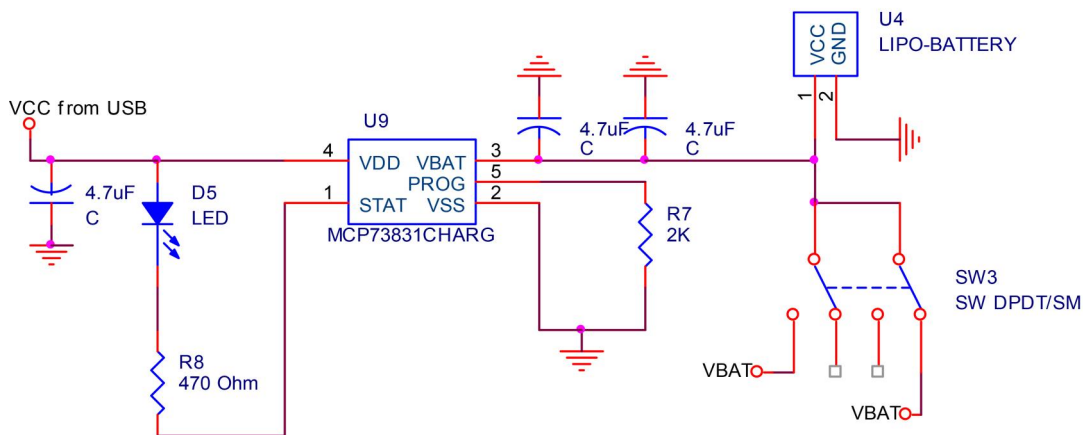
The system includes a lithium-ion/lithium-polymer battery charger, enabling the user to charge the battery *via* USB power. The Li-Po battery is charge controlled by an MCP73831.73831 (2024), a highly advanced linear charge management controller which is also suitable for space-limited and cost-sensitive applications. The charger provides a charge algorithm for Li-Po battery cell with constant current followed by constant voltage. Typical charge profiles are given in the MCP73831 Datasheet (2020). The charging circuit in the implemented design is given in Figure 5(b).

Debugging and testing initial design

The system was initially designed on a breadboard using a Microchip Microstick for 5 V PIC24 K Series Devices (PIC24K. 2024) with PIC24FV32KA302 devices attached. Microstick for 5 V PIC24 K Series Devices provides an easy-to-use, economic development environment for 16-bit microcontrollers, offering an integrated debugger/programmer, a socket and pin-out for the target microcontroller chip to facilitate flexible development. An ESP8266 8-pin Wi-Fi



(a) Schematic design of DS18B20 temperature sensor connection with PIC24FV16KA301



(b) Battery charging circuit

Figure 5. Schematic diagrams of temperature sensor and battery charging circuit's connections to the microcontroller unit.

module is attached to the initial circuit for debugging. Figure 6(a) illustrates the initial circuit.

The ESP8266 8-pin Wi-Fi module is configured with AT commands by connecting to a PC *via* FTDI 3.3 V Basic Breakout USB to Serial adapter (FTDI (presenter) 2024). The same method was used to debug serial communication between the PIC microcontroller and Wi-Fi module.

Implementing and printing the circuit

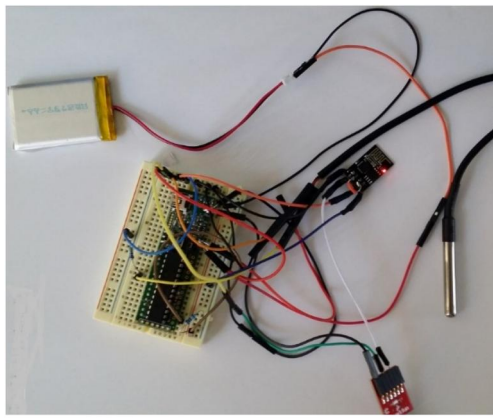
Once the initial breadboard circuit was debugged and tested, the confirmed version of the circuit was implemented on a Veroboard, allowing the PIC microcontroller to be programmed *via* the PICKIT3 debugger/

programmer. As a more feasible device for the application, a 20-pin PIC24FV32KA301 is used for the Veroboard design. All of the system components except the battery charging circuit are implemented and tested on a Veroboard design. Figure 6(b) shows the circuit on the Veroboard.

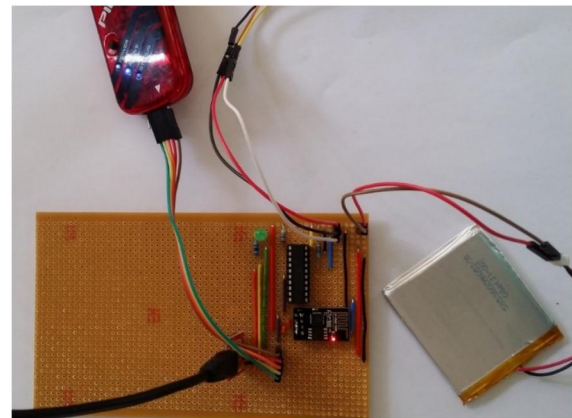
The final version of the system is on a printed circuit board including a Li-Po battery charging circuit *via* USB power. The final circuit (schematic shown in [Figure 7](#)) uses a 20-pin PIC24FV16KA301 microcontroller.

Results and discussion

Dhivya et al. (2024) reported a heat stress management system based on environmental sensors.



(a) Initial system design with Microstick for 5V PIC24 K on a breadboard



(b) Veroboard circuit design with PIC24FV32KA301

Figure 6. Prototypes of the smart vest circuit design.

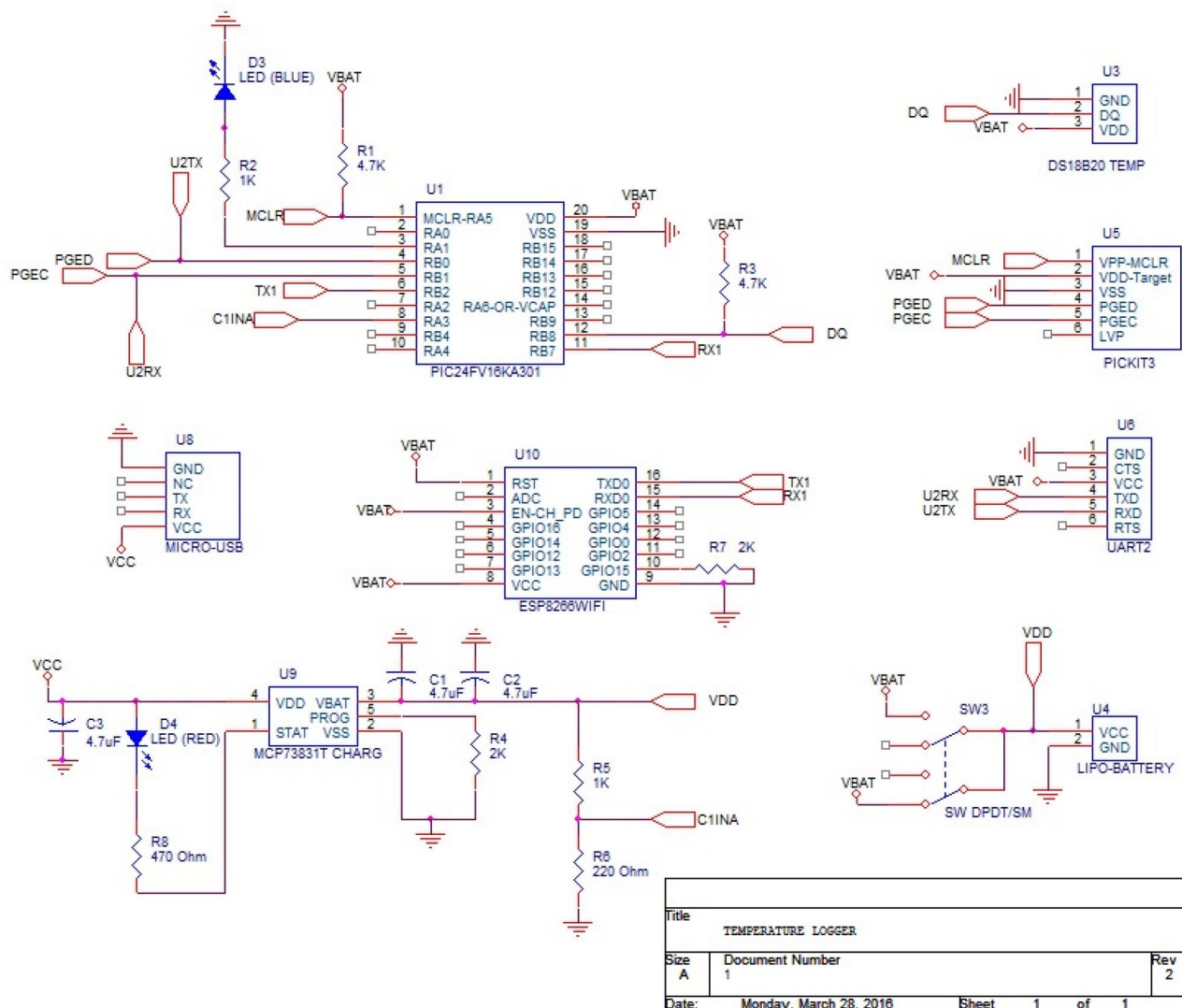


Figure 7. Completed version of circuit schematics.

However, research reported that use of ambient temperature as an inaccurate measure compared to body temperature in assessing heat stress and particularly

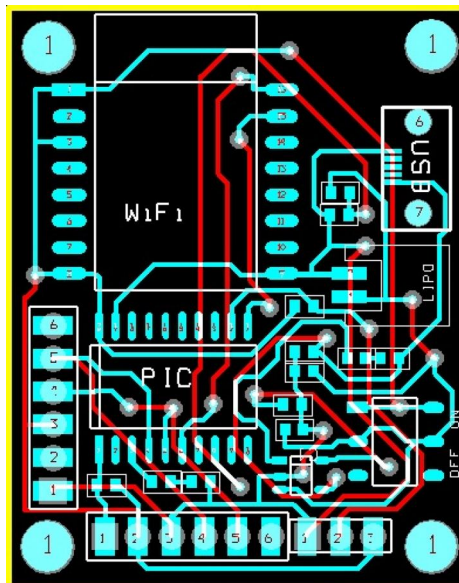
clothing has been identified as a key factor in heat transfer between humans and their surroundings (Qian and Fan 2006). Some early technological

solutions developed for heat stress management of construction workers using wearable clothing are at prototype stage of scientific research or are non IoT enabled (Edirisinghe and Blismas 2015). Kim et al. (2020) developing a system using a commercially available arm band and for heat stress management of construction workers. While arm band limits the ability to accurately investigate the microclimate conditions (thermal characteristics) and impact due to clothing layer, which is a significant factor in assessing heat stress (Edirisinghe and Jadhav 2017). Rigorous investigation of accuracy through microclimate variations in heat stress studies under varying operational set-ups to understand the thermal comfort of the wearer is crucial in designing a wearable tailored for construction workers due to specific working conditions in construction. Moreover, some physiological measurements (Kim et al. 2020) were not captured in real time and is disruptive to regular work. A tailor made solution developed specifically for construction industry (such as use of PPE) is the unique

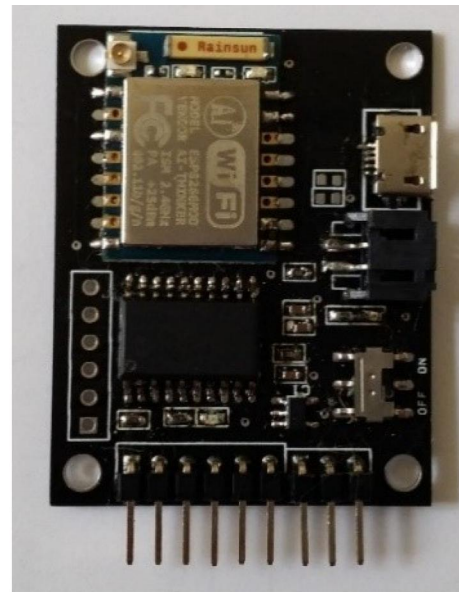
contribution of this research. In addition, the existing research lack the theoretical underpinnings of systematic reflective process in translational research. This research is guided by the methodology of solving problems using a reflective process, guided by the Technology Readiness Levels (TRLs) proposed by Mankins (1995), an approach widely accepted by scholars as applied research translational principals and adopted in practice. The smart vest development stage reported in this paper together with test results are discussed below.

Functional testing

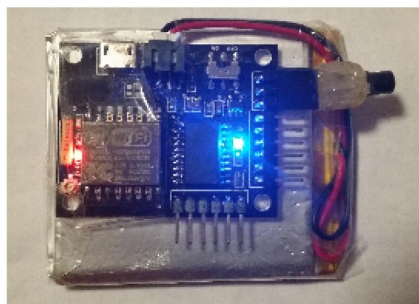
The PCB layout with component footprints and connection routes is given in Figure 8(a). The complete system has been debugged and tested for its expected functionality. The temperature reader, Wi-Fi and battery charger sub-modules are integrated into the printed circuit board design and the components are attached to achieve complete system functionality.



(a) printed circuit board layout



(b) PCB with components soldered



(c) final IoT-enabled circuitry



(d) IoT-enabled smart vest used in field

Figure 8. Circuit design and PCB used in testing.

The final completed printed circuit board (PCB) is shown in [Figure 8\(b\)](#).

The functional testing revealed that system uploads data to the webserver with a 16-second lag, since the server needs this time interval between each data item added to the channel. Given that the system works at this efficiency, the 2,000mAh Li-Po battery allows the device to be utilised for 42h without power-saving modes being activated. The charging circuit has been tested for its functionality and the time for charging the 2,000mAh single cell Li-Po battery is estimated to be four hours. The system has also been tested for Wi-Fi connectivity, demonstrating the ability to reconnect and resume functionality if a worker goes out of range and then re-enters the Wi-Fi network area. Also, the system is automatically resumed if there is a temporary Wi-Fi network failure. [Figure 8\(c\)](#) shows the final system.

Battery life and further laboratory testing

Power consumption depends on both the microcontroller performance and the code running on the microcontroller. Power can be continuously drawn from the 2,000 mAh Li-Po battery for 42h without any power-saving modes activated with data uploaded at a frequency of once every 16s. The charging circuit has been tested for functionality for charging the battery. The current state-of-the-art design with the same data upload frequency lasts only 22h.

Further laboratory tests were conducted to assess the system's accuracy and responsiveness, utilising a conditioning oven (model: Incubator M250-RH). The testing involved temperatures ranging from 30 to 45° Celsius with increments of 0.5°. During these tests, the system exhibited an error range of -0.4375 to $+0.06$ ° Celsius, indicating its ability to maintain close alignment with the desired temperature settings.

Smart vest: circuit tailored for the safety vest

The complete system design was iteratively developed and tested based on industry feedback. The ultimate aim of the proposed system is to provide an IoT-based smart vest tailored for measuring the microclimate and skin temperature of construction workers. The revised system was repeatedly tested with a frequency of data upload of 16s. The waterproof temperature sensor maintains accuracy even in wet conditions, such as sweating. The sensor is accurate to ± 0.5 °C and has a resolution of 0.0625 °C, allowing

the system to identify small and rapid temperature variations.

An enclosure was designed to ensure wearer comfort in collaboration with construction industry contributors and experts in fashion and textiles, leading to an air mesh fabric with a cushioning effect being incorporated. Features such as easy attachment to and detachment from the garment and water resistance, considering the metabolic conditions of the wearers in summer (such as sweat), were also taken into account. [Figure 8\(d\)](#) shows the final version of the circuit attached to the smart vest in operation of which further details are reported elsewhere.

Implications

Globally, heat stress presents a significant occupational health challenge, affecting construction workers. Exposure to extremely hot environments can result in heat stress, placing workers at risk of illnesses such as heatstroke, heat exhaustion and heat cramps (Edirisinghe and Blismas 2015). These conditions, which occur when the body becomes unable to control its temperature, can even cause death or permanent disability. Most outdoor workers and worker in confined spaces are at risk of heat stress, as are workers in hot environments (Borg et al. 2021; De Lima et al. 2021). The exploration of the 'Digital Skin' concept (Edirisinghe 2019) on smart construction sites along with smart construction workers wearing smart PPE and the emergence of Construction 5.0 underscore the transformative impact of technology on the construction industry to address these issues using digital innovation including wearable IoT based PPE. The smart vest circuit presented in this paper helps enable wider adoption in the industry as an integral to IoT based heat stress management systems (Kodikara Edirisinghe and Gunathilake 2024) for realising the vision of Construction 5.0. Hence, the findings of this study has the potential to enable step change in providing tailor made IoT based PPE for the construction sector.

Conclusions

The literature on wearable technology and the IoT illuminates limitations requiring attention for the wider adoption such as privacy concerns (Poongodi et al. 2020) and need for comprehensive assessment and predictors (Niknejad et al. 2020) for consumer acceptance.

The proposed system offers significant technological and design advantages. The incorporation of a waterproof temperature sensor ensures accurate readings, even in wet conditions, with high precision and a resolution capable of detecting small and rapid temperature variations. Unlike readily available smart wristwatches, which tend to malfunction in harsh environments, the proposed system allows construction workers to focus entirely on their tasks while continuously monitoring their health. Additionally, the system's capability to communicate with web-servers and other devices facilitates both local and global access to temperature data, thereby enhancing its utility. The ease of USB charging further augments its usability.

Moreover, the system's cost and energy-efficient design, utilizing readily available components, renders it commercially viable for both individual and corporate use. This innovative and cost-effective solution provides essential protection against heat stress for construction workers, ensuring their safety and well-being.

In future iterations, the system will be augmented with additional health-monitoring features, such as heart rate variability-based monitoring, environmental humidity and galvanic skin response to offer a more comprehensive health assessment for construction workers. These enhancements will be implemented with careful consideration of privacy and consumer acceptance, ensuring that the system remains both effective and user-friendly. Beyond these sensor improvements, the system holds potential for broader applications in industries with high safety risks, such as manufacturing, mining, and agriculture. Moreover, the integration of predictive analytics could enable real-time decision-making for better workforce management and safety protocols. Such advancements would provide even more value to industries seeking to protect their workers while maximizing operational efficiency.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability

The data that support the findings of this study are available on request from the corresponding author.

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